



PNP-BV, revisited

Forward PDE simulation for ORR selectivity — and using AI for scientific computing

■ LIVE DEMO

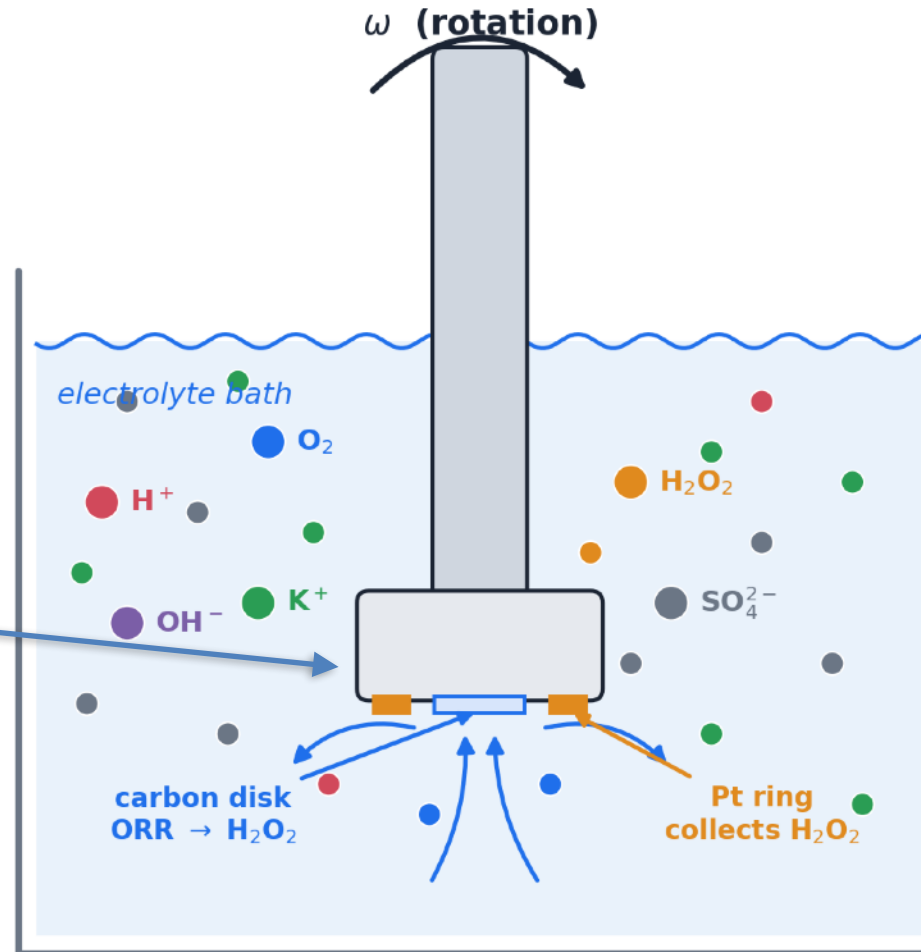
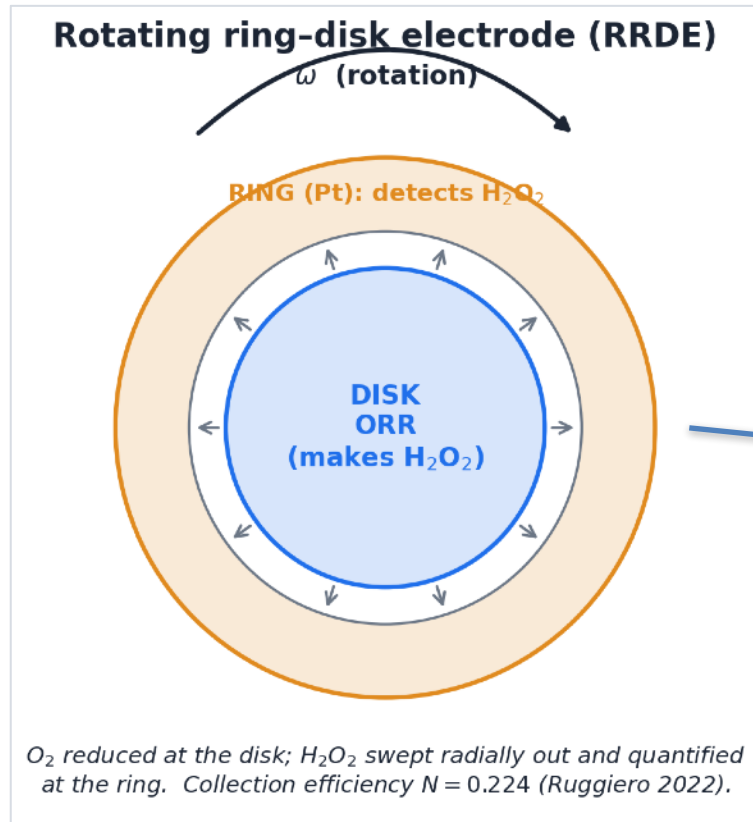
Using claude code to re-derive

An autonomous nonlinear toy problem for structural unidentifiability:
three polynomial Lotka–Volterra systems, same observed prey

■ What I'll cover

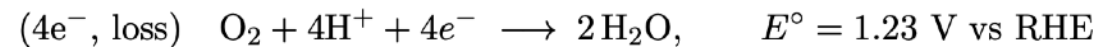
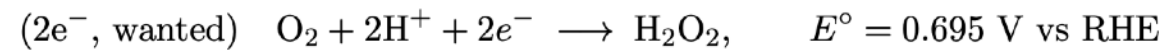
- 1 The problem — ORR selectivity
- 2 The model — PNP + Butler–Volmer + steric correction
- 3 Why it's numerically hard
- 4 How we solve it
- 5 Prior work I'm building on
- 6 Verification
- 7 What's next — toward the real chemistry

The RRDE cell: rotating ring-disk electrode in the electrolyte bath

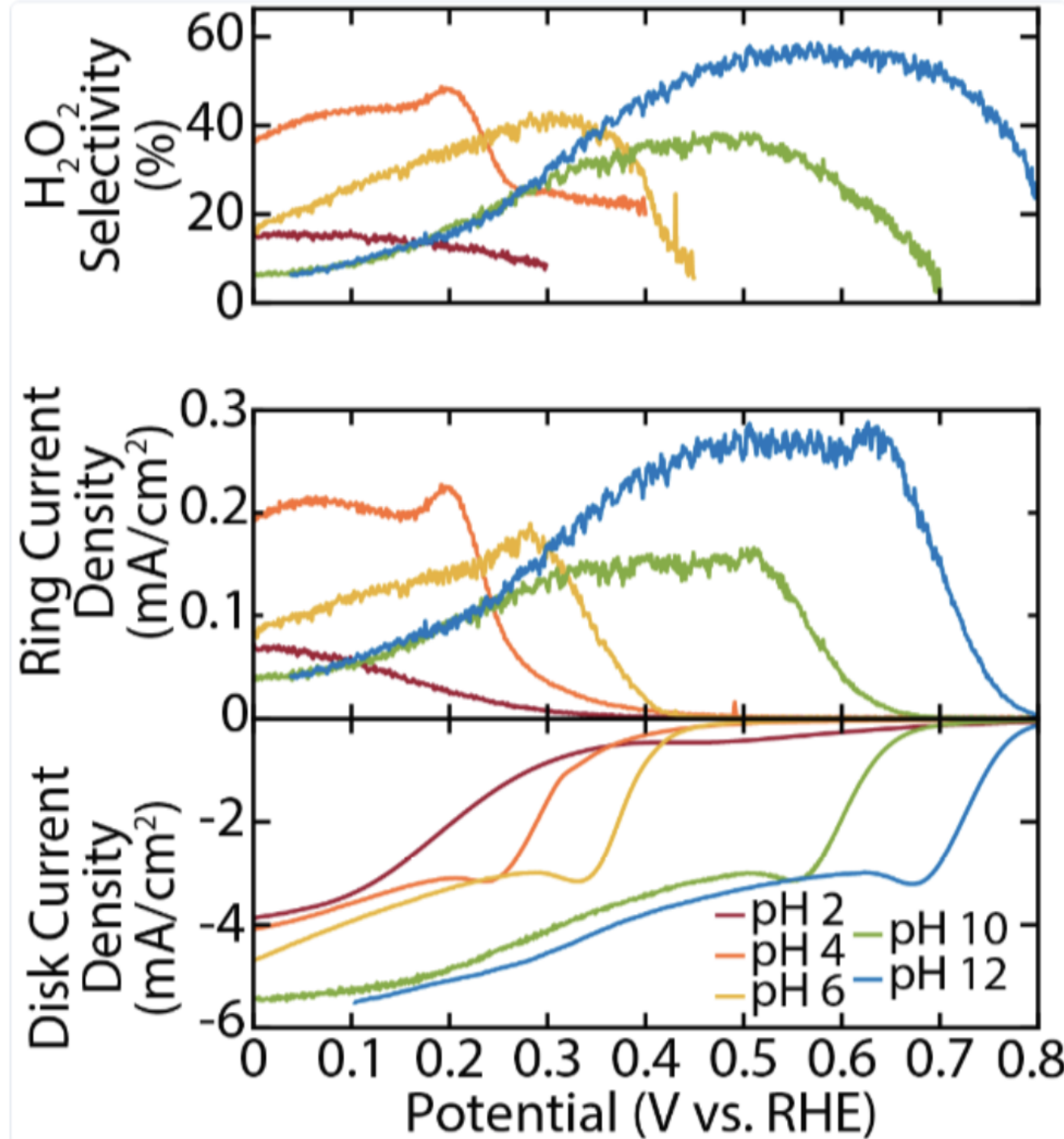
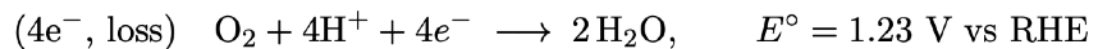
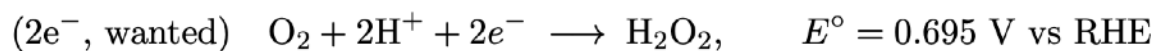


Species in the bath

- O_2
dissolved reactant (O_2 -saturated)
- H^+
acidic medium; ORR co-reactant
- H_2O_2
 $2e^-$ product (swept to the ring)
- K^+
support cation; $\text{M}^+ = \text{Li}/\text{Na}/\text{K}/\text{Cs}$
- SO_4^{2-}
support anion ($\text{K}_2\text{SO}_4 + \text{H}_2\text{SO}_4$)
- OH^-
from water / KOH (minor, acidic)
- H_2O
solvent (the bath itself)



- **H2O2 Selectivity** → the ratio of the peroxide reaction to the water reaction
- **Disk Current density** → the net rate of both reactions combined.
- **Ring current density** → how much of the peroxide pathway is firing: the disk makes H₂O₂, the ring downstream detects it.



■ Goals

- Better understand the mechanisms underlying the data
- Pin down kinetic parameters
- Create a more accurate model for the underlying chemistry
- Determine under which conditions selectivity is best to facilitate easier manufacturing

The model: PNP + Butler-Volmer + Steric correction

- **PNP in the diffuse layer:** ion diffusion coupled to the potential field, so charged species also electromigrate.

$$\frac{\partial c_i}{\partial t} = \nabla \cdot \left(D_i \left[\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi \right] \right) \quad -\nabla \cdot (\varepsilon \nabla \varphi) = \sum_i z_i F c_i.$$

- **Butler–Volmer flux at the electrode:** reaction rate scales with exp(overpotential), tied to the applied voltage.

$$R_r = k_{0,r} \left[c_O e^{-\alpha_r F \eta_r / RT} - c_R e^{(1-\alpha_r) F \eta_r / RT} \right], \quad \eta_r = \varphi_m - \varphi_s - E_{,r}^\circ.$$

- **Steric Correction** removes very high, non-physical ion concentrations

$$\mu_i^{\text{steric}} = -\ln \theta, \quad \theta = 1 - \sum_i a_i c_i,$$

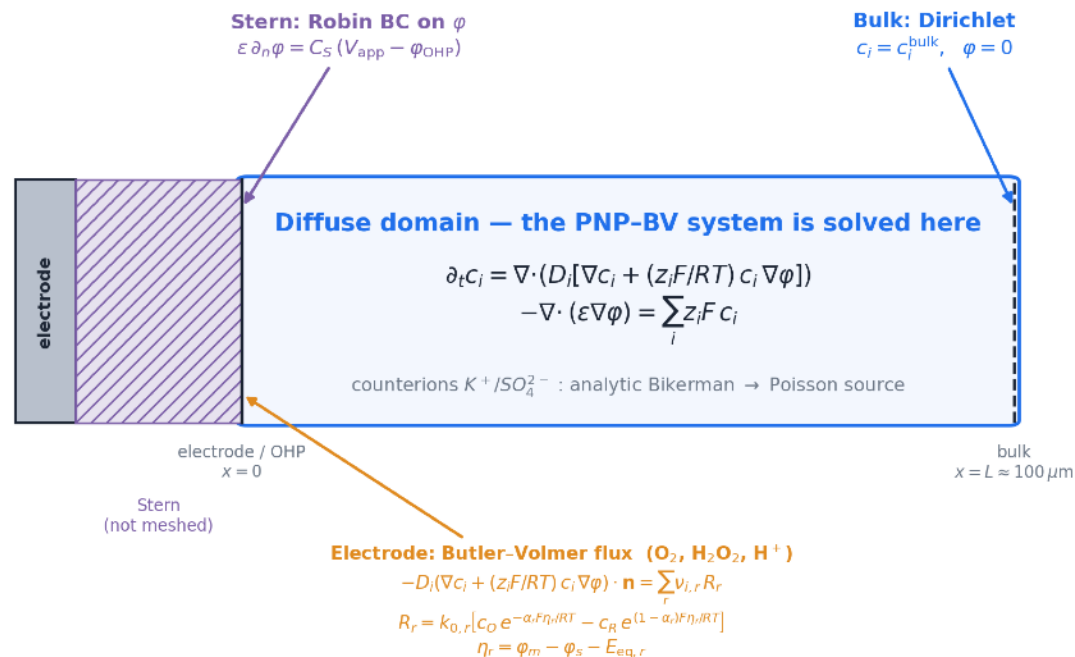
- **Stern layer** enters as a Robin boundary condition (next slide).

$$\varepsilon \partial_n \varphi = C_S (V_{\text{app}} - \varphi_{\text{OHP}}),$$

■ Model setup

- **Electrode** (left) → Stern layer → diffuse PNP region → bulk (right).
- **Stern layer:** ions pack against the electrode with no room to move, modeled as a capacitor (Bazant 2001) — the voltage drop $V - \phi(\text{OHP})$ sets a Robin boundary condition on ϕ .
- **Bulk:** fixed concentrations; potential pinned to ground to set the drop across the domain.
- **Domain** $\approx 100 \mu\text{m}$ (l_{eff}); changing it changes the results.

The full model on the domain: governing equations + every boundary condition



Why it's numerically hard

- **Concentrations span ~10 orders of magnitude** across the domain.
- **The potential varies sharply** right at the Stern layer.
- **The exponential Butler–Volmer term** drives steep buildup in c and ϕ at the boundary.
- **Diffusion and electromigration** act on very different timescales — the system is stiff.

$$c_i \approx c_i^{\text{bulk}} \exp(-z_i F \phi / RT)$$

How we solve it

1 Solve $\mu = \ln(c) + z^*\phi$ — mirrors the no flux analytical solution

$$\mu_H = u_H + \frac{z_H F}{RT} \varphi$$

2 Log-rate Butler–Volmer — build the rate in log-space, exponentiate once — avoids overflow / underflow. PUT EQ

$$R_r = k_{0,r} \left[c_O e^{-\alpha_r F \eta_r / RT} - c_R e^{(1-\alpha_r) F \eta_r / RT} \right] \longrightarrow \begin{aligned} \log r_r^{(\text{cat})} &= \ln k_{0,r} + u_O - \alpha_r \frac{F \eta_r}{RT}, & r_r^{(\text{cat})} &= \exp(\log r_r^{(\text{cat})}), \\ \log r_r^{(\text{anod})} &= \ln k_{0,r} + u_R + (1 - \alpha_r) \frac{F \eta_r}{RT}, & r_r^{(\text{anod})} &= \exp(\log r_r^{(\text{anod})}), \\ R_r &= r_r^{(\text{cat})} - r_r^{(\text{anod})}, & \eta_r &= \varphi_m - \varphi_s - E_{r,\circ} \end{aligned}$$

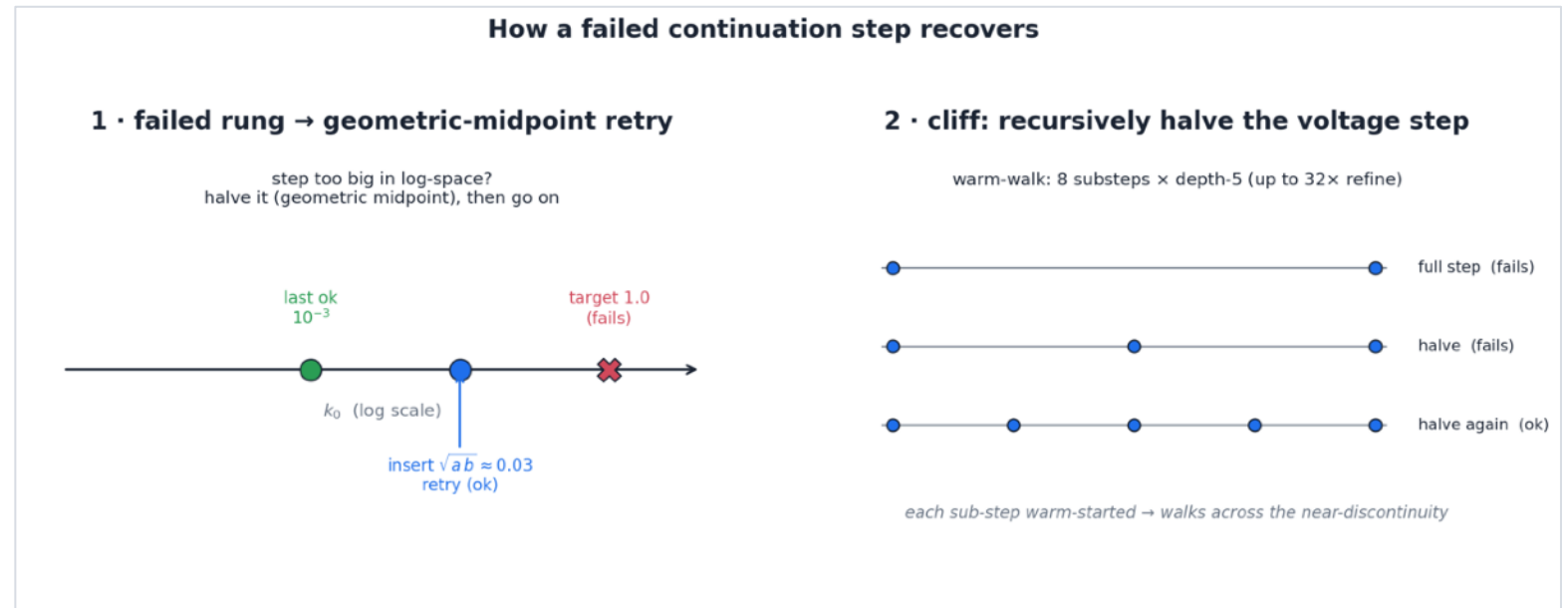
3 Analytic closure for non-reactive ions

$$c_b(x) = c_b^{\text{bulk}} \frac{(1 - \sum_{j \in \text{dyn}} a_j c_j(x)) e^{-z_b F \varphi / RT}}{\theta_b + a_b c_b^{\text{bulk}} e^{-z_b F \varphi / RT}}, \quad \theta_b = 1 - \sum_j a_j c_j^{\text{bulk}} - a_b c_b^{\text{bulk}}.$$

4 Continuation — never solve the hard problem cold (next slide).

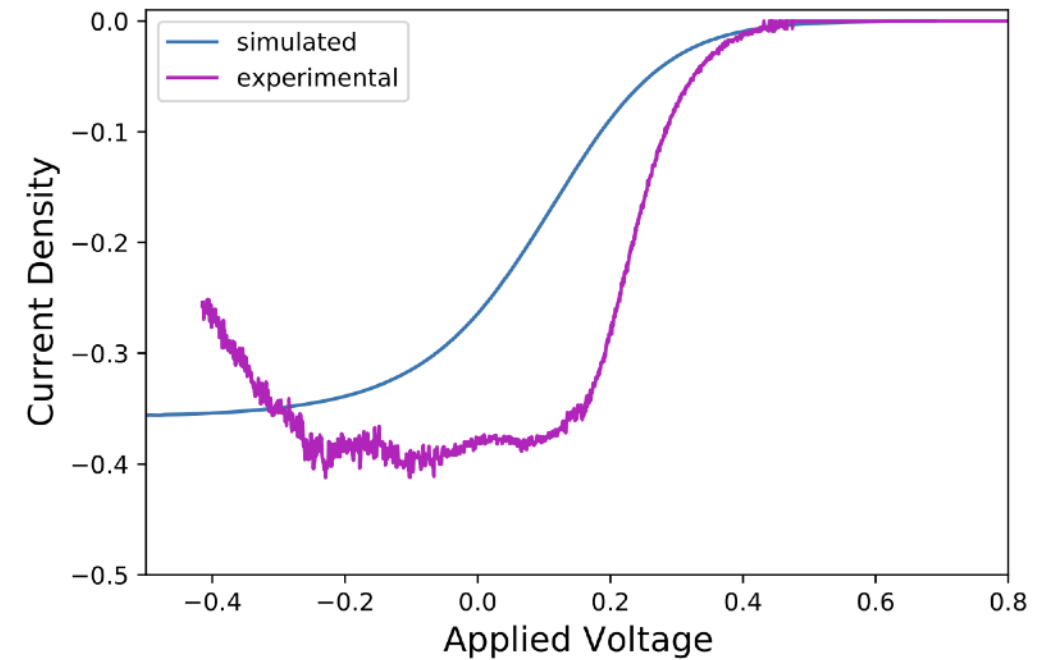
■ Continuation: deform easy → hard

- **Reaction rate k_0 :** from $\sim 10^{-12} \times$ physical up to physical, warm-starting each step.
- **Stern capacitance:** ramp up (stiffer as C_S grows).
- **Voltage walk:** converge near the equilibrium voltage, then step to every other voltage.
- **Recursive bisection:** insert intermediate voltages when a step fails.



Prior work:

- **Earlier group modeling** didn't solve the full PDE.
- **Same analytic closure** — but applied to all ions (not just non-reactive ones) — plus an ODE for the potential.
- **Butler–Volmer closed** by matching boundary O_2 to the BV rate with a root-finder.
- **A reduced model**, not a PDE solve.

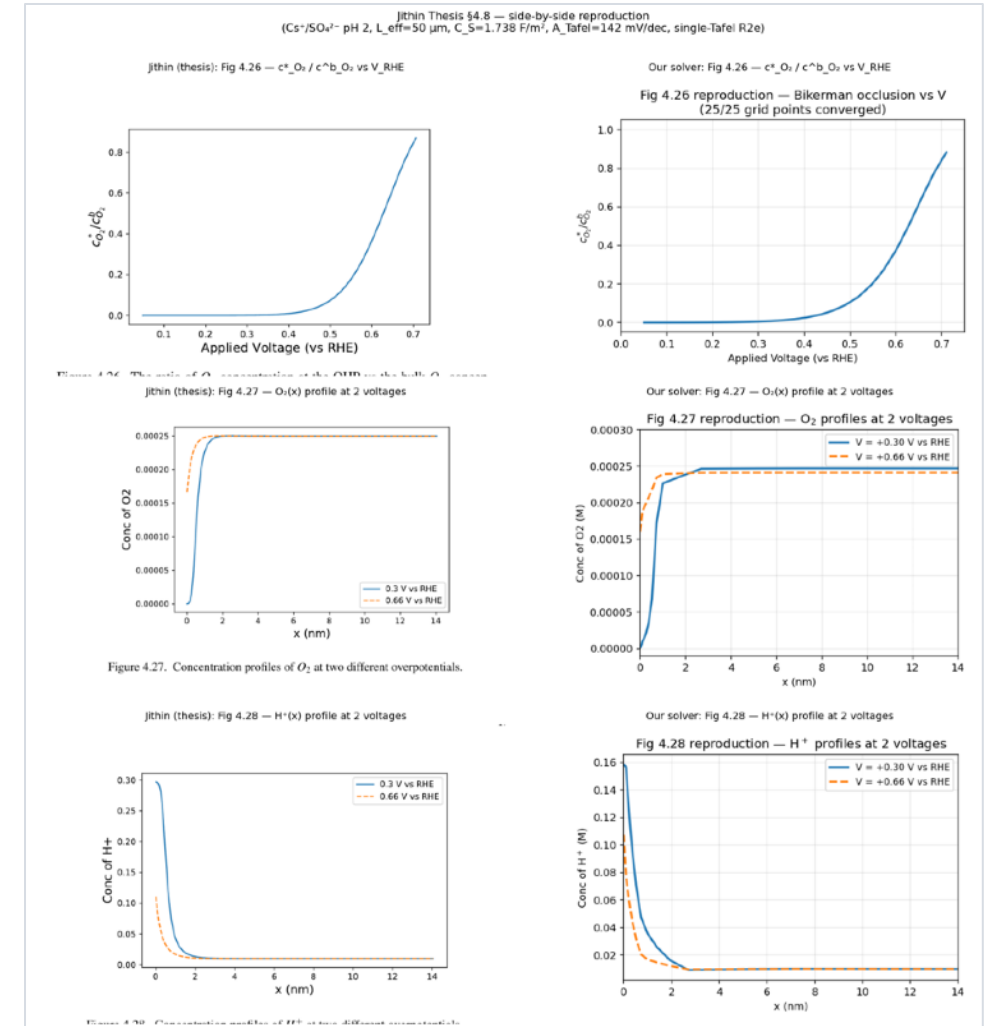


■ Limitations of the closure approach

- **One reaction only ($2e^-$):** the $4e^-$ reduction to water is missing, makes modeling selectivity impossible
- **Not a transport solve:** adding or changing physics is hard. The model relies on closed-form tricks valid only under specific conditions.

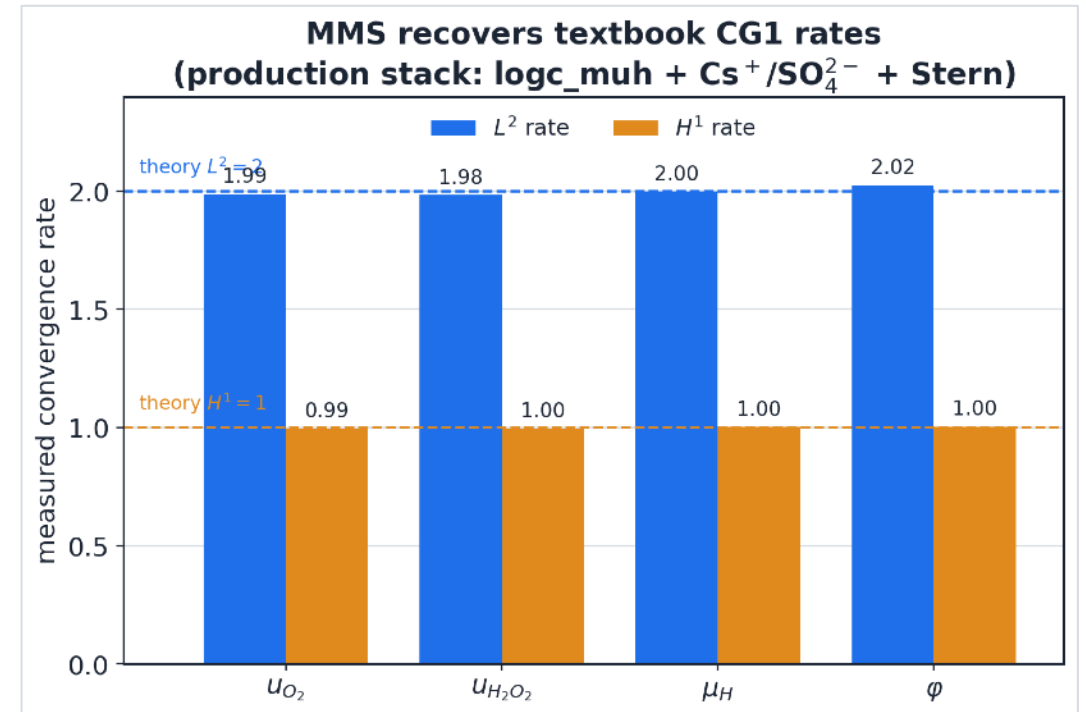
Verification: agreement with prior work

- **Matched the prior setup:** single $2e^-$ reaction, same diffusivities and steric coefficients.
- **Prior thesis figures (left)** vs this solver's reconstruction (right).
- **Result:** O_2 -ratio and profiles agree to ~5%.



Verification: manufactured solutions

- **Method of manufactured solutions:** insert a known exact solution, derive the source term it implies, check the solver recovers it.
- **Observed rates match CG1 theory:** $L^2 \approx 2$, $H^1 \approx 1$ for every field.



What's next?

Refining the model until we get a good fit to the physical data.